

Drones Fail Silently

- **Safety critical drone deployments are growing**
Larger drones flying near humans in military, surveillance, & agriculture
- **No onboard or realtime detection**
16% of crashes caused by imperfect flight checks ([Grindley et al., 2024](#))
- **Propeller deterioration goes unnoticed**
Wear and tear like tiny bends accrue over time until larger failures



On-Device Propeller Anomaly Detection

Run a quantised audio ML model on an Arduino Nano 33 BLE Sense.
Prior works used large autoencoders. Classify propeller health in real time.



< 256 KB RAM

fits Arduino SRAM budget



F1 > 0.9

on held-out test set



< 50 ms

inference latency target

Pipeline & Milestones

1

Audio Input

TII UAE dataset (normal, 1 broken, 2 broken, 3+ broken)

2

STFT / Mel Spectrogram

frequency feature extraction

3

1D-CNN / Autoencoder

Lower dimensionality

4

INT8 Quantisation

Arduino deployment (with EON compiler)

5 pts

Model fits on Hardware RAM/Flash

Accuracy > 80%+ fits on Arduino Nano 33 BLE Sense

10 pts

Deployed inference while on laptop

F1 > 0.9 and running live on board

15 pts

Deployed inference on real drone

Normal vs. broken propellor tests on Holybro X500

What Success Looks Like

● 4-class classification

Normal, 1-Broken, 2-Broken, 3/4-Broken

● Model size

Target < 200 KB Flash after INT8 quantisation

● Accuracy

F1 > 0.9 on TII UAE held-out test split

● Baseline comparison

Versus existing CNN from literature ([Katta et al., 2022](#))